

Paper A-06: Ecological Energy and Trophic Networks: A Constraint-Driven Flux Dynamics (CDFD) Perspective

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August 2026

Shared Part IV notation and evidence boundaries are defined in `PART_IV_METHODS_AND_CLAIM_BOUNDARY.md`. This paper states only its local observables, comparison, and failure conditions.

Abstract

This paper develops a falsifiable CDFD/AFL model of Earth Ecological Energy Transport. It maps solar energy, nutrients, biomass production, and consumer demand to driving flux Φ , stoichiometry, trophic bottlenecks, habitat limits, and loss to active constraint C , functional diversity, migration, storage, and trophic rerouting to responsiveness S , and community composition, soil state, disturbance, and evolutionary legacy to structural memory M_s . The target outcome is productivity, respiration, transfer efficiency, and recovery, tested against flux towers, food-web observations, isotope tracers, productivity, and respiration records. The paper treats universality as a shared model architecture rather than an assertion that unlike systems have the same material mechanism. It states the negative result that would make the mapping unnecessary and places the domain claim inside the cumulative CDFD programme and its reproducible runtime boundary.

Consolidation Scope and Provenance

This consolidated paper retains the ecological energy-transport and trophic-network scopes formerly carried by Papers A-06 and D-08. It distinguishes ecosystem productivity and transfer efficiency from trophic-cascade outcomes, and requires field-native ecological baselines for either task.

1 Introduction

An ecosystem is a biological engine. It captures high-grade solar energy via photosynthesis and dissipates it as heat, passing the intermediate chemical energy up through successive trophic

levels. The efficiency of this energy transfer is notoriously low (the 10% rule), meaning the higher trophic levels are inherently fragile, operating at the absolute limit of available flux.

In CDFD, the food web is not a static list of species; it is a highly active, constraint-bound transport network. Extinction and population booms are simply the system's attempt to regulate its internal fluid pressure.

2 The Ecological Adaptive Surface

We govern the trophic network using the Adaptive Operating Ratio:

$$\Psi_s = \left(\frac{\Phi}{C} \right) \cdot S \cdot M_s \quad (1)$$

For an ecosystem (e.g., a coral reef or a savannah):

- **Flux** (Φ): The caloric energy transfer (biomass consumed) moving from prey to predator.
- **Constraint** (C): The carrying capacity of the environment, including resource scarcity, habitat boundaries, and metabolic friction (heat loss).
- **Responsiveness** (S): The biodiversity and phenotypic plasticity of the ecosystem. A highly diverse ecosystem can quickly reroute energy flow if one specific prey species declines.
- **Memory** (M_s): Niche specialization and symbiotic dependencies. Over millions of years, species become hyper-specialized (high memory). While highly efficient in stable environments, this locks the topological routing of energy into rigid, fragile pathways.

2.1 Trophic Cascades and Keystone Species

A stable ecosystem balances at $\Psi_s \approx 1$. The caloric flux Φ perfectly supports the existing constraints and biomass.

Consider the removal of a keystone species (e.g., apex predators like wolves). Without the downward predatory constraint, the herbivore population (flux Φ) explodes. The system enters an over-flux regime ($\Psi_s \gg 1$). The herbivores rapidly consume all available primary producers, completely destroying the foundational responsiveness (S) of the environment. The adaptive ratio then crashes violently to $\Psi_s \ll 1$, resulting in a mass die-off.

Conversely, consider an ecosystem locked by high memory M_s (hyper-specialization, like Pandas relying solely on bamboo). If the bamboo is wiped out, the specific energy flux Φ drops to zero. Because the system's memory is locked, it cannot adapt to a new food source (cannot adjust C or S). The isolated node faces immediate necrosis (extinction).

3 Measurement Layer

4 Cross-Domain Model Upgrade: Mechanism, Scale, and Tests

4.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Earth Ecological Energy Transport, the proposed drive is solar energy, nutrients, biomass production, and consumer demand. The active limitation is stoichiometry, trophic bottlenecks, habitat limits, and loss. Responsiveness is carried by functional diversity, migration, storage, and trophic rerouting, while retained state is represented by community composition, soil state, disturbance, and evolutionary legacy. The outcome to explain is productivity, respiration, transfer efficiency, and recovery.

Table 1: Operational CDFD/AFL map for Paper A-06.

Term	Paper-specific observable meaning	Measurement question
Φ	solar energy, nutrients, biomass production, and consumer demand	What rate, load, gradient, or demand enters the focal system?
C	stoichiometry, trophic bottlenecks, habitat limits, and loss	Which measured bottleneck limits transfer, function, or recovery?
S	functional diversity, migration, storage, and trophic rerouting	Which process changes effective capacity or reroutes the load?
M_s	community composition, soil state, disturbance, and evolutionary legacy	Which retained state makes equal present loads produce different futures?
Y	productivity, respiration, transfer efficiency, and recovery	Which outcome is predicted out of sample, with uncertainty?

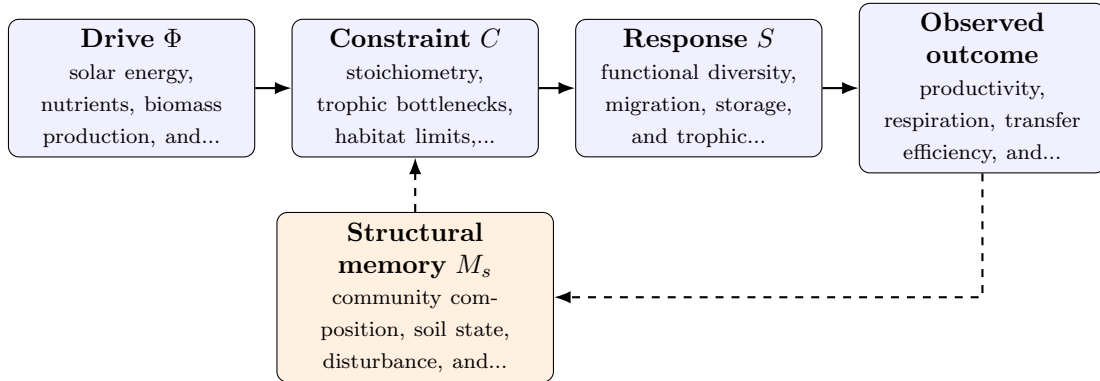


Figure 1: Paper A-06 observable CDFD/AFL architecture for Earth Ecological Energy Transport. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

4.2 Paper-Specific Causal Hypothesis

For this paper, the hypothesized causal sequence is specific. A change in solar energy, nutrients, biomass production, and consumer demand arrives faster than functional diversity, migration, storage, and trophic rerouting can compensate. This raises or redistributes stoichiometry, trophic bottlenecks, habitat limits, and loss. If the load persists, the retained state encoded

by community composition, soil state, disturbance, and evolutionary legacy changes the next response, so a later event of equal magnitude can produce a different trajectory in productivity, respiration, transfer efficiency, and recovery. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

4.3 Cross-Scale Translation Without Mechanistic Erasure

For the Earth systems family, the cross-domain translation compares coupled transport, finite environmental capacity, buffering, and retained landscape or ecosystem state. It cannot erase conservation laws, geometry, or scale; it must expose where those domain equations create comparable overload and recovery structures. [2, 3, 1] The required scale declaration for this paper is seconds to centuries; organisms to biomes. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

4.4 Evidence Ladder and Discriminating Tests

Table 2: Evidence ladder for Paper A-06. Each rung can fail independently.

Test	Design	Discriminating result
Proxy audit	Measure flux towers, food-web observations, isotope tracers, productivity, and respiration records and preregister how each record maps to Φ , C , S , M_s , and Y .	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe a resource pulse followed by drought, harvest, or consumer loss while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper’s stated falsifier is: the proposed constraint-memory terms do not explain transfer efficiency or recovery.

4.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from flux towers, food-web observations, isotope tracers, productivity, and respiration records and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of productivity, respiration, transfer efficiency, and recovery. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the cross-domain translation audit.

5 Conclusion

Ecosystems are macroscopic thermodynamic circuits. Biodiversity is not merely an aesthetic preference; it is the critical Responsiveness (S) parameter that prevents topological gridlock. By applying CDFD, conservation biology gains a unified physics framework: preserving nature requires maintaining the mathematical elasticity of its energy transport networks.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Part-specific runtime diagnostics are under each Part's `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: `Part_E_Synthesis/outputs/`.

Figures: `Part_E_Synthesis/figures/`.

6 Reference Layer

References

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